

Aditya Sugandhi

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SUMMARY

Full-stack engineer with 5+ years building **AI-native products from zero to production**. Hands-on experience designing **agentic AI systems, multi-agent orchestration, RAG pipelines with vector databases**, and **knowledge-graph-driven architectures**. Ship end-to-end across **Python/FastAPI** backends and **React/Next.js/TypeScript** frontends. Thrive in founding-team environments—own architecture, build infrastructure, iterate fast.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript/JavaScript, SQL, Go, Java, Scala, Bash, C++
- **AI & Knowledge Systems:** Agentic AI, Multi-Agent Orchestration (LangChain, CrewAI), RAG Pipelines, LLMs (GPT, Llama, Groq), Vector DBs (Pinecone, ChromaDB, pgvector), Knowledge Graphs, Semantic Search, Prompt Engineering
- **Frontend:** React, Next.js 14, TypeScript, TailwindCSS, Zustand, Framer Motion, Vapi Voice AI
- **Backend & Infra:** FastAPI, Node.js, PostgreSQL, Redis, MongoDB, Kafka, Spark, GraphQL, WebSockets, AWS (EKS, Lambda, S3), Kubernetes, Docker, CI/CD (GitHub Actions), Vercel, Grafana

WORK EXPERIENCE

Software Engineer — AI & Full-Stack Systems

Nov 2024 – Present

Florida State University

Tallahassee, FL

- Architected **agentic AI system** with **multi-agent orchestration**: autonomous agents discover, classify, and map **35K+ devices** across **172+ networks** using **LLM-driven reasoning** and **graph-based dependency resolution**, reducing manual investigation by **85%**.
- Built **RAG pipelines** with **vector DB integration** (pgvector, ChromaDB) and **knowledge-graph traversal** for context-aware retrieval; agents reason over structured relationships, cutting debugging time by **40%** for **15+ teams**.
- Delivered **full-stack platform**: **FastAPI** backend (**120+ REST endpoints** + **WebSocket streaming**), **React/Next.js** frontend with real-time dashboards, **Redis caching**—owned architecture from schema to UI as sole engineer.

Software Engineer — Data & AI Platform

Jan 2023 – Aug 2024

Prof. Olmo Zavala Romero — FSU

Tallahassee, FL

- Designed **vector embedding pipelines** and **semantic search** using Transformers; deployed retrieval engine processing **1,000+ documents** at **95% accuracy**, **sub-200ms latency**—foundational architecture decisions made independently.
- Built **distributed data pipelines** with **Python, Spark, Kubernetes**; orchestrated parallel execution achieving **+40% throughput** over **1TB+ datasets** with **99% success rate**.
- Partnered with researchers, data scientists, and platform engineers to define **AI pipeline standards**; reduced **pipeline runtime** by **50%** through architecture optimization.

Software Engineer — Full-Stack & Integrations

Oct 2020 – Jul 2022

Aspire Systems

Chennai, India

- Built **event-driven microservices** with **Kafka** processing **500K+ events/day**; designed fault-tolerant async workflows handling **10x traffic spikes** with **99.5% availability**.
- Developed **React/TypeScript frontends** and **REST/GraphQL APIs** end-to-end across **3 product lines**; mentored **5 engineers** on best practices, code review, and TDD.

PROJECTS

- **AI Resume Optimization Platform (Private)** — **Multi-agent system**: job scanning, semantic matching, and optimization agents with **RAG pipelines**, **vector DB integration**, **LLM tool-calling**. Tech: **Python, TypeScript, Next.js, FastAPI, Llama 3.3, Groq API**.
- **Enterprise Slack Bot — Agentic AI (Internal)** — **Agentic chatbot** with **tool/agent integrations** for automated workflow management and retrieval-augmented responses. **24/7 uptime**, **sub-40s latency**. Tech: **Python, Node.js, AWS EC2, OAuth 2.0, WebSockets**.
- **Pragle Chiropractic — Production Client Website Live** — Shipped production website for healthcare client: **Vapi Voice AI** integration, multi-step booking system (**Zod validation**), symptom questionnaire, **content-driven architecture** via typed JSON config. Tech: **Next.js 14, TypeScript, TailwindCSS, Framer Motion, Zustand, Vercel**.

EDUCATION

Florida State University

M.S. in Computer Science

Tallahassee, FL

- **Relevant Coursework:** Distributed Systems, Advanced Database Systems, Machine Learning, Data Communications, Algorithms

SRM Institute of Science & Technology

B.Tech. in Computer Science

Chennai, India

PUBLICATIONS & AWARDS

- Review Classification & False Feedback Detection (IJAST)
- 2nd Place, SRM University Hackathon (2019)
- AWS Solutions Architect Course (Udemy, 2024)